

AERO70038 - Problem set 4

Design and Trim

1. You are asked to design a rotor for a large helicopter whose primary mission is to lift logs and heavy construction equipment in the mountains. The gross weight is 100,000 lbs, rotor diameter is 120 ft, with 7 blades, average operating altitude of 15,000 ft and a rotor tip speed of 650 ft/s.
 - (a) Calculate the tip chord and equivalent rotor solidity.
 - (b) How much power would be required to hover out of ground effect at the design condition?

Choose and obtain data on an appropriate aerofoil.

(a) design rotor with best efficiency at hovering

∴ design rotor with maximum PL at hovering

for optimum hovering rotor, $b_r = \text{constant} = b_{\text{tip}}$

$$\therefore b_e = 3 \int_0^r b_{\text{tip}} r dr = \frac{3}{2} b_{\text{tip}}$$

$$\therefore C_{P_o} = \frac{1}{2} \int_0^r b_{\text{tip}} C_{d_o} r^2 dr = \frac{1}{6} b_{\text{tip}} C_{d_o} = \frac{1}{9} b_e C_{d_o}$$

$$P_L = \frac{T}{P} \quad \therefore P_{L_{\max}} = \left(\frac{P}{T} \right)_{\min}$$

$$\text{where } \frac{P}{T} = \frac{C_P}{C_T} \Omega R = \left(K \sqrt{\frac{C_T}{2}} + \frac{1}{9} b_e \frac{C_{d_o}}{C_T} \right) \Omega R$$

$$\therefore \text{at } \frac{d}{dC_T} \left(\frac{P}{T} \right) = 0, \quad C_T \text{ for } P_{L_{\max}}$$

$$\therefore \frac{1}{2} K \frac{1}{\sqrt{2} C_T} - \frac{1}{9} 6e \frac{C_{d0}}{C_T^2} = 0$$

$$\therefore \frac{\sqrt{2}}{4} K C_T^{\frac{3}{2}} - \frac{1}{9} 6e C_{d0} = 0$$

$$\therefore C_T = \left(\frac{2\sqrt{2}}{9} \frac{6e C_{d0}}{K} \right)^{\frac{2}{3}}$$

$$\therefore C_T = \frac{W}{\rho A (\Omega R)^2} = 0.01396$$

$\therefore 6e$ can be found as C_{d0} and K is known

$$\therefore C_{tip} = \frac{6_{tip} \pi R}{N_b} = \frac{2}{3} \frac{6e \pi R}{N_b}$$

$$b) C_p = K \frac{C_T^{\frac{3}{2}}}{\sqrt{2}} + \frac{1}{9} 6e C_{d0}$$

$$P = \rho A (\Omega R)^3 C_p$$

accounting for ground effect ...

2. In the design of a swept tip rotor blade, it is desired to introduce only as much sweep angle as necessary to keep the Mach number incident to the leading edge of the blade equal to a constant, say M_c , over the tip region. Because of unsteady flow effects, the critical azimuth angle for the design is taken to be at $\psi = 100^\circ$.
- Find an expression for the sweep angle as a function of r , the hover tip Mach number $M_{\Omega R}$, the design Mach number M_c and the advance ratio μ .
 - If $M_{\Omega R} = 0.65$ and $M_c = 0.8$, plot the sweep angle required as a function of radial blade station for a series of advance ratios.

(a) At hover, $(U_T)_H = \Omega y$

$$\therefore M_{\Omega R} = \frac{(U_{T, tip})_H}{a} = \frac{\Omega R}{a}$$

$$\therefore M_H = \frac{\Omega y}{a} = \frac{\Omega R}{a} \frac{y}{R} = M_{\Omega R} r$$

At forward flight,

$$\begin{aligned} M &= M_H + M_{\infty} = M_{\Omega R} r + \frac{V_{\infty} \cos \alpha \sin \psi}{a} \\ &= M_{\Omega R} r + \frac{1}{a} \mu (\Omega R) \sin 100^\circ = M_{\Omega R} r + M_{\Omega R} \mu \sin 100^\circ \end{aligned}$$

$$\therefore M \cos \Lambda = M_c$$

$$\therefore \Lambda = \cos^{-1} \left(\frac{M_c}{M} \right) = \cos^{-1} \left[\frac{M_c}{M_{NR} (r + \mu \sin 100^\circ)} \right]$$

(b) see solution

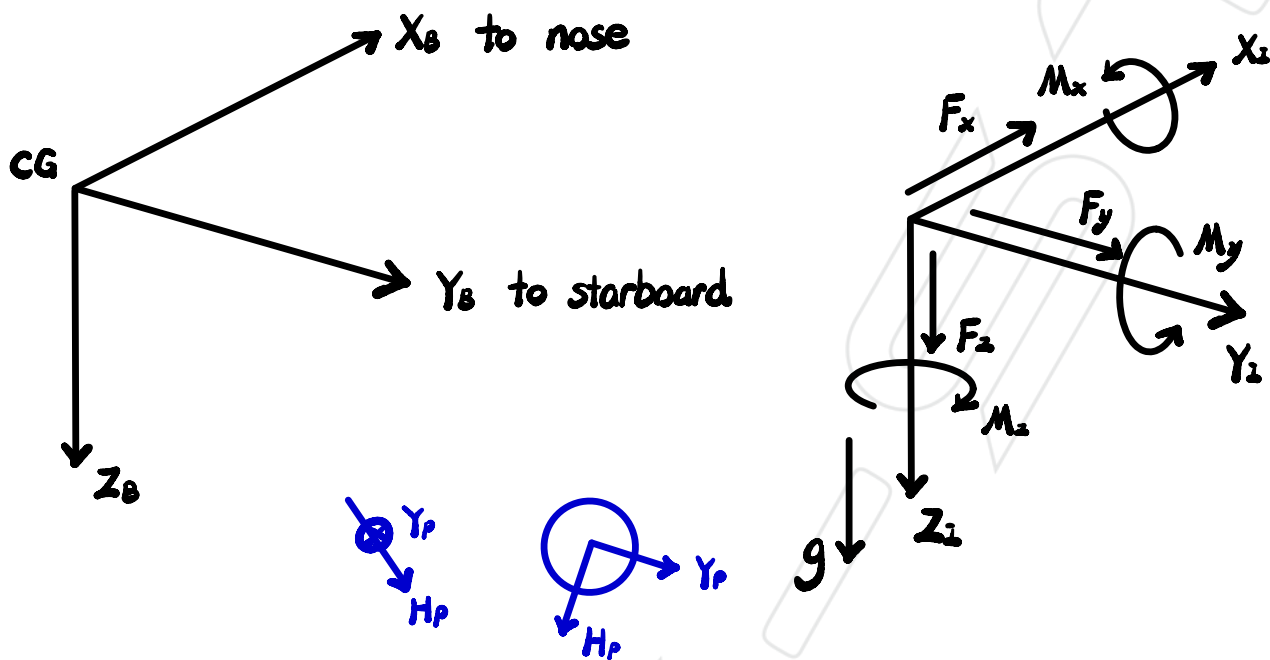
3. The Sikorsky-Boeing SB-1 Defiant is a technology demonstrator that was proposed to replace the Sikorsky UH-60 Black Hawk. It is a compound helicopter with two coaxial rotors and a pusher propeller (see Figure 1).

Write the force and moment balance equations for this aircraft. You may assume that the centre of gravity is aligned with the shaft for simplicity.



Figure 1: Sikorsky-Boeing Defiant flight demonstrator.

Assume CG align with shaft for simplicity, the axis notation as shown below :



Therefore, assume $\theta_{FP1} = \theta_{FP2} = \theta_{FP}$, propeller thrust parallel to nose

$$F_z = W - (T_{MR1} + T_{MR2}) \cos \theta_F \cos \phi_F + D \sin \theta_{FP} - (H_{MR1} + H_{MR2}) \sin \theta_F \\ + (Y_{MR1} + Y_{MR2} + Y_F + Y_H) \sin \phi_F - T_P \sin \theta_F = 0$$

$$F_x = (T_{MR1} + T_{MR2}) \sin \theta_F \cos \phi_F - D \cos \theta_{FP} - (H_{MR1} + H_{MR2}) \cos \theta_F \\ + T_P \cos \theta_F = 0$$

$$F_y = (T_{MR1} + T_{MR2}) \cos \theta_F \sin \phi_F + (Y_F + Y_{MR1} + Y_{MR2} + Y_H) \cos \phi_F = 0$$

About hub of the bottom rotor.

$$M_x = M_{xMR1} + M_{xMR2} + M_{xF} + (W \sin \phi_F + Y_F) h - Q_P - Y_{MR2} h_{MR2} \\ + Y_H h_T = 0$$

$$M_y = M_{yMR1} + M_{yMR2} + M_{yF} + (W \sin \theta_F - D \cos \alpha_s) h = 0$$

$$Q_{MR1} = \frac{1}{T} Y_H = 0 \dots \text{no information about transmission system}$$

can further simplified by small angle approximation

Answers:

1. (a) $c_{tip} = 6.6 \text{ ft}$, $\sigma_e = 0.367$
(b) $P = 15,545.5 \text{ HP}$
2. (a) $\Lambda = \cos^{-1} \left(\frac{M_c}{M_{\Omega R}(r + \mu \sin 100^\circ)} \right)$
(b) Plot
3. Set of equations.